Computer

Class 9 (English Medium)

Solved Exercises — Sindh Textbook Board

7 lessons — 204 questions

ڪمپيوٽر جي بنيادي چاڻ

[1-1] Which device converts a paper (hard) copy into a soft copy inside the computer?

(a) Printer

(b) Plotter

✓ (c) Scanner

(d) Barcode reader

[1-2] Which memory is lost when the power goes off?

(a) It is permanent

✓ (b) It is lost as soon as the power goes off

(c) It processes extra storage

(d) It is looked after through hardware

[1-3] What kind of software is a media player?

(a) Business software

(b) Educational software

✓ (c) Entertainment software

(d) Productivity software

[1-4] Which program supervises all the work of the computer?

✓ (a) Operating system

(b) Utility program

(c) Language translator

(d) Device driver

[1-5] Which tool do modern (high-level) languages use for translation?

✓ (a) Compiler

(b) Interpreter

(c) Converter

(d) Assembler

[1-6] What is the set of wires that joins the CPU to the main memory called?

(a) Control bus

✓ (b) Data bus

(c) Address bus

(d) Memory bus

[1-7] Which computers are the most expensive of all?

✓ (a) Super computer

(b) Mainframe computer

(c) Mini computer

(d) Micro computer

[1-8] Without what can a computer not run?

✓ (a) Operating system

(b) Utility program

(c) Device driver

(d) Business software

[1-9] In which generation did the Graphical User Interface (GUI) first appear?

(a) First generation

✓ (b) Fourth generation

(c) Mechanical era

(d) Electro-mechanical era

[1-10] Who is the person expert in writing and running programs?

(a) Database administrator

(b) Web designer

✓ (c) Software engineer

(d) Graphic designer

Question: [1-11] Write about the use of the computer in any two fields of life.

Answer: In education, the computer is used for teaching and learning, school management and online learning; in health, doctors use computers for patients' records, tests and guidance in treatment. In the same way, in banking, business, agriculture and communication, the computer makes everyday work easier.

Question: [1-12] State the difference between a compiler and an assembler.

Answer: A compiler translates a whole program written in a high-level language into machine language in one go; an assembler translates the instructions of a program written in assembly language into machine language.

Question: [1-13] State the difference between hardware and software.

Answer: Hardware is the visible, touchable physical parts of the computer, such as the motherboard, keyboard and monitor; software is the set of instructions that makes the computer work, such as the operating system and application programs. Without hardware, software cannot run, and without software, hardware is useless.

Question: [1-14] Explain impact and non-impact printers with examples.

Answer: An impact printer prints by striking a ribbon onto the paper, like a dot-matrix printer; a non-impact printer prints without striking, like an inkjet or laser printer, which give clean and quiet printing.

Question: [1-15] State the difference in the use of a hard disk, a USB flash disk and an SD card.

Answer: A hard disk is the large-capacity permanent storage fitted inside the computer; a USB flash disk is a small, portable storage used to move data from one computer to another; an SD card is a tiny memory card

mostly used in mobiles, cameras and other small devices.

Question: [1-16] Which is better for a school, a CRT or a flat-panel monitor, and why?

Answer: For a school a flat-panel monitor is better, because it takes less space, uses less electricity and shows a clear picture, while a CRT monitor is heavy and bulky.

Question: [1-17] State the difference between system software and application software.

Answer: System software runs the computer's hardware and all its activity, such as the operating system and drivers; application software does some particular job of the user, such as writing a document, doing calculations or playing a game. Without system software, application software cannot run.

Question: [1-18] Write the names of any five parts found on the motherboard.

Answer: On the motherboard there are the microprocessor (CPU), RAM, ROM, ports, slots, buses, BIOS chip, connectors, transistors and diodes.

Question: [1-19] Complete the table of the five generations of computers: write the period, technology and example of each generation.

Answer: First generation: 1940–1956, vacuum tube, examples ENIAC and UNIVAC; second generation: 1956–1965, transistor, examples IBM 7094 and IBM 1401; third generation: 1964–1971, integrated circuit (IC), examples IBM 360 and IBM 370; fourth generation: 1971 to today, microprocessor, examples IBM PC and Macintosh; fifth generation: today onwards, artificial intelligence (AI), examples robots and expert systems.

Question: [1-20] Give three examples each of business, education, entertainment and productivity software.

Answer: Business: banking software, accounting software, inventory

management; education: e-learning programs, school management software, learning programs; entertainment: media player, computer game, video program; productivity: word processor (documents), spreadsheet, presentation program.

Question: [1-21] Which statement matches the ALU?

Answer: Solving arithmetic and logical calculations — the ALU does calculations such as addition, subtraction, multiplication, division and comparison.

Question: [1-22] Which statement matches an input device?

Answer: Hardware and a tool that enters data into the computer — such as the keyboard, mouse and scanner.

Question: [1-23] Which statement matches a secondary storage device?

Answer: Large storage capacity and permanent state — secondary storage keeps data saved permanently, such as a hard disk.

Question: [1-24] Which statement matches productivity software?

Answer: Helping with spreadsheets and databases — productivity software helps with documents, spreadsheets and presentations.

Question: [1-25] Which statement matches the mechanical era?

Answer: The era of hand-operated machines — early machines were run by hand, such as the abacus.

Question: [1-26] Which statement matches a digital computer?

Answer: Processing using the binary system — a digital computer works on 0 and 1 (binary).

Question: [1-27] Which statement matches a micro computer?

Answer: PC — a micro computer is called a personal computer (PC), which is for the personal use of one person.

Question: [1-28] Make a timeline of the major changes in the evolution of the computer.

Answer: Mechanical era: abacus, Napier's bones, slide rule, Pascaline and Leibniz's calculator; electro-mechanical era: Hollerith's tabulating machine; electronic era: first generation (vacuum tube), second generation (transistor), third generation (integrated circuit), fourth generation (microprocessor), fifth generation (artificial intelligence).

Question: [1-29] Make a list of input and output devices and write their functions.

Answer: Input devices: keyboard (entering writing), mouse (pointing), scanner (entering images), microphone (entering sound); output devices: monitor (showing the result), printer (printing on paper), speaker (producing sound), projector (showing on a big screen).

Question: [1-30] List the important features of a computer.

Answer: The computer works fast and accurately on given instructions, stores a large amount of data, repeats the same work without getting tired, and can do many kinds of work in different fields.

Question: [1-31] Give examples of analogue, digital and hybrid computers around you.

Answer: Analogue: speedometer, voltmeter; digital: desktop computer, laptop, mobile phone; hybrid: hospital measuring instruments and petrol pump machines, which convert analogue measurement into digital numbers.

Question: [1-32] Label the given block diagram with names: which parts are inside and outside the CPU?

Answer: Left side: input devices (data comes in); right side: output devices (the result goes out); inside the CPU: memory (holding data), control unit (directing all parts), ALU (doing calculations).

آپريٽنگ سسٽم جي بنيادي چاڻ

[2-1] Which software carries out the basic operations of the computer?

(a) Antivirus

(b) Start menu

(c) Office automation

✓ (d) Operating system

[2-2] Which program makes contact with the computer system possible?

(a) My Documents

(b) Start menu

(c) Desktop

✓ (d) Interface

[2-3] What is the process of checking the computer and preparing the operating system for work called?

(a) Resource management

✓ (b) Booting

(c) Error checking

(d) File management

[2-4] With how many users can a multi-user operating system work at the same time?

(a) One

(b) Two

✓ (c) Many

(d) Only organised

[2-5] What is the transfer of data between the processor and memory and the control of resources called?

✓ (a) Resource

(b) File

(c) I/O

(d) User

[2-6] Which operating system divides similar jobs into groups?

✓ (a) Batch processing

(b) Time sharing

(c) Single user

(d) Real time

[2-7] Which system makes sure that work is finished within the fixed time without delay?

(a) Batch processing

✓ (b) Hard real time

(c) Soft real time

(d) Time sharing

[2-8] DOS is an example of what?

(a) Real-time operating system

(b) Multiprocessing OS

✓ (c) Single-user OS

(d) Multi-user OS

[2-9] What is the group of programs that helps in the daily work of an office called?

(a) Operating system

(b) Email

✓ (c) Office automation

(d) Antivirus

[2-10] Which operating system finishes work within the fixed time?

(a) Batch processing

✓ (b) Real-time processing

(c) Single user

(d) Time sharing

Question: [2-11] Define the operating system and write its examples.

Answer: The operating system is the program that runs the computer and manages its files, memory, input/output and external devices. Examples: Windows, DOS, Linux, Android, Mac and iOS.

Question: [2-12] State the important functions of the operating system.

Answer: The functions of the operating system are: booting (preparing the system), resource management (distribution of resources), user interface (contact with the user), memory management, input/output management, file management, device management and user management.

Question: [2-13] Explain the difference between GUI and CLI.

Answer: In a GUI there are pictures, icons and menus, and work is done with mouse clicks, such as Windows and iOS; in a CLI, commands have to be typed and remembered, such as MS-DOS and Unix. A GUI is easy for the ordinary user; a CLI is fast but harder to learn.

Question: [2-14] Which resources does the operating system manage in resource management?

Answer: In resource management, the operating system manages the distribution (allocation), return (de-allocation) and shared use (shared memory) of hardware and software resources such as the processor, memory, input/output devices and files.

Question: [2-15] State the difference between single-user and multi-user operating systems, and tell which is better for an institution and why.

Answer: In a single-user system, only one user can work at a time, such as DOS and Windows 98; in a multi-user system, many users can work at the same time, such as Unix and Linux. For an institution, multi-user is better, because many people can work at the same time and resources are

shared.

Question: [2-16] What is the purpose of office automation software? How are its programs linked with each other? Explain with examples.

Answer: The purpose of office automation is to make the daily work of an office easy, fast and better. Its programs are linked with each other: for example, an Excel table can be brought into a Word document, and email can be sent through Outlook.

Question: [2-17] Which programs are in MS Office? State the use of each one.

Answer: MS Office includes Word (writing documents), Excel (spreadsheets and calculations), PowerPoint (presentations), Access (database management) and Outlook (email and contacts).

Question: [2-18] What is the difference between soft real time and hard real time?

Answer: In hard real time, the work must be finished within the fixed time and there is no room for delay, such as a traffic control system; in soft real time, the work should be finished within the time but a small delay can be tolerated, such as online transactions and multimedia.

Question: [2-19] Why is it necessary to install antivirus software?

Answer: Antivirus protects the computer from viruses, keeps data and the system safe from harm, and keeping its virus database updated also protects against new viruses.

Question: [2-20] Which statement matches CLI?

Answer: Interface — CLI is a written-commands interface, in which the user types commands to give instructions to the computer.

Question: [2-21] Which statement matches Windows?

Answer: Operating system — Windows is a well-known operating system.

Question: [2-22] Which statement matches a process?

Answer: The work the system is doing — a process is the work that the system is doing at that time.

Question: [2-23] Which statement matches MS Excel?

Answer: Office automation — MS Excel is an office automation program, used for spreadsheets and calculations.

Question: [2-24] Which statement matches a real-time operating system? (As per the remaining pairing in the book's table)

Answer: Time-sharing operating system — both are types of operating systems.

Question: [2-25] Which statement matches input/output management?

Answer: Function of the operating system — input/output management is one of the functions of the operating system.

آفیس آٹومیشن

[3-1] Which chart would best show the shares of three business partners?

(a) Line chart

(b) Bar chart

(c) Area chart

✓ (d) Pie chart

[3-2] Which software is used for calculations?

(a) MS-Word

✓ (b) MS-Excel

(c) ASPPT

(d) MS-Access

[3-3] Which sign is used to start a formula in Excel?

(a) /

✓ (b) =

(c) —

(d) *

[3-4] What is the software for formatting a document called?

✓ (a) MS-Word

(b) MS-PPT

(c) MS-Excel

(d) MS-Access

[3-5] Which function puts data in a particular order?

(a) Filtering

✓ (b) Sorting

(c) Organising

(d) Grouping

[3-6] What is used to change the overall design of a Word document?

✓ (a) Themes

(b) PHP

(c) Watermark

(d) Margins

[3-7] Which is the correct formula to find the total marks in the given table?

(a) =B3+C2+B2+E2

(b) =B2+C2

Question: [3-8] Which is the correct formula to find the percentage in the given table? [Note: there is an error in the printed options — (a) and (c) are printed identically and no option is mathematically correct; the correct formula is given below]

Answer: The correct formula is =(B2+C2+D2+E2)/400*100 — the obtained marks are divided by the total 400 and multiplied by 100. Of the printed options, the (a)/(c) one, which divides by 400, is the intended answer.

[3-9] What are charts basically used for?

(a) Showing analytical data

(b) Arranging data

(c) Arranging data in different columns

✓ (d) Showing data in picture form

[3-10] What is the most important thing for making a table of contents effective?

(a) Title

✓ (b) Page numbering

(c) Page order

(d) Index

Question: [3-11] What is a word processor? Write two of its uses.

Answer: A word processor is software with which documents are written, corrected and formatted. Uses: writing letters and documents; making documents beautiful by adding pictures, tables and colours.

Question: [3-12] Explain the difference between the two kinds of orientation.

Answer: In portrait, the page is taller than it is wide (upright page); in landscape, the page is wider than it is tall (sideways page).

Question: [3-13] How are margins set in a document?

Answer: In the Page Layout tab, click Margins and choose a ready-made margin, or write the top, bottom, left and right margin sizes yourself in Custom Margins.

Question: [3-14] Discuss the margin and paper size options in the Page Setup group.

Answer: Margins fix the empty space (border) around the page; Size chooses the size of the page, such as A4 or Letter. Both options are found in the Page Setup group.

Question: [3-15] Explain the different breaks in MS Word.

Answer: With a break, a new page or a new column is started in the document. A page break starts a new page, and a column break starts the writing in a new column.

Question: [3-16] State the difference between Send to Back and Bring to Front.

Answer: Bring Forward brings the selected object in front of the other objects; Send Backward sends the selected object behind the other objects. Both are used to order objects lying on top of each other.

Question: [3-17] Explain the Columns option in the Page Setup group.

Answer: The Columns option divides the writing into two or more columns,

as in newspapers and magazines.

Question: [3-18] State the Align Position and Wrap Text options of the Arrange group.

Answer: Position places a picture or object at a fixed place on the page; Wrap Text arranges the writing around the picture.

Question: [3-19] In what way does a Table of Contents (ToC) help the reader?

Answer: A table of contents holds a list of headings and their pages, so the reader can quickly find the wanted topic and reach that page directly.

Question: [3-20] What must be written in the MS Excel formula bar to multiply 37 by 15?

Answer: =37*15 — start with the equals sign, write 37, the multiplication sign (*) and 15, then press Enter; the answer 555 will come.

Question: [3-21] What is a watermark in a document? Give its examples.

Answer: A watermark is faint writing or a picture appearing behind the writing on the page, which shows the status or ownership of the document. Examples: DRAFT, CONFIDENTIAL, a company logo.

Question: [3-22] Give examples of the use of spreadsheets in business.

Answer: In business, spreadsheets are used to keep accounts, budgets, salaries, stock and sales records, and profit-and-loss calculations.

Question: [3-23] Which statement matches Wrap Text?

Answer: Arranging the writing around the picture — with Wrap Text, the writing wraps around the picture.

Question: [3-24] Which statement matches an automatic table of contents?

Answer: It is made on the basis of headings — an automatic table is made by itself from the headings of the document.

Question: [3-25] Which statement matches the Page Setup group?

Answer: For margins, paper size and setting — margins, page size and orientation are fixed from the Page Setup group.

Question: [3-26] Which statement matches the use of Excel?

Answer: Organising, displaying, formatting and calculating data — Excel organises, displays, formats and calculates data.

Question: [3-27] Which statement matches charts?

Answer: Giving data a pictorial expression — a chart shows data in picture form.

Question: [3-28] Which statement matches the common use of MS Word?

Answer: For documentation — MS Word is commonly used for making documents.

Question: [3-29] From the given cricket table, count the runs from each kind of hit: 1s=21, 2s=7, 3s=0, 4s=14, 5s=0, 6s=9.

Answer: One-run hits: 21×1=21; two-run hits: 7×2=14; three-run hits: 0×3=0; fours: 14×4=56; five-run hits: 0×5=0; sixes: 9×6=54. The hits give 145 runs in total; the remaining 4 runs are extras, because the total score is 149.

Question: [3-30] From the given cricket data, find the player's strike rate: runs 149, balls 64.

Answer: Strike rate = (149 ÷ 64) × 100 = 232.81.

Question: [3-31] Find the separate share of each kind of hit in the 149 runs. (The question is read as each hit-type's share)

Answer: One-run hits: 21/149×100=14.09%; two-run hits: 14/149×100=9.40%; fours: 56/149×100=37.58%; sixes: 54/149×100=36.24%; three- and five-run hits: 0%.

ڈیٹا کمیونیکیشن کمپیوٹر نیٹورکس

[4-1] What is wired media called?

(a) Targeted media

(b) Directed media

✓ (c) Guided media

(d) Unguided media

[4-2] Of how many parts is a communication system made?

(a) Three parts

(b) Four parts

✓ (c) Five parts

(d) Six parts

[4-3] What are both the physical and logical addresses?

(a) Separate

✓ (b) Unique

(c) Permanent

(d) Temporary

[4-4] Which is the organisation of electrical and electronics engineers?

✓ (a) IEEE

(b) IETF

(c) ITU

(d) ANSI

[4-5] Which topology is it in which all computers are joined to a central hub?

(a) Bus

✓ (b) Star

(c) Ring

(d) Tree

[4-6] What is it called when the shape of the signal changes between sender and receiver?

(a) Attenuation

(b) Interference

(c) Noise

✓ (d) Distortion

[4-7] By which route does a router send data packets?

✓ (a) Shortest

(b) Largest

(c) Cheapest

(d) Easiest

[4-8] What is changing a digital signal into an analogue signal called?

✓ (a) Modulation

(b) Modification

(c) Sendert

(d) Sliplancing

[4-9] How many bits are used in an IPv4 address?

(a) 16

✓ (b) 32

(c) 64

(d) 128

[4-10] What is the weakening of a signal during transmission called?

(a) Jitter

(b) Distortion

(c) Noise

✓ (d) Attenuation

Question: [4-11] Write the qualities of a good communication system and

explain any one of them.

Answer: A good communication system has three basic qualities: delivery, accuracy and timeliness. Delivery means that the data must reach the correct destination, not some other place.

Question: [4-12] Explaining with an example of one part, describe the parts of a communication system.

Answer: A communication system has five parts: message (the information sent), sender (the device that sends the message), receiver (the device that receives the message), medium (the path, such as a wire or radio waves) and protocol (the rules). Example: in sending an email, the writing is the message, the computers are the sender and receiver, the internet is the medium, and the email rules are the protocol.

Question: [4-13] State the work of these network devices: amplifier, router, switch, hub.

Answer: An amplifier strengthens a weak signal; a router joins two or more networks and chooses the best route for data; a switch joins the devices of a network (computers, printers) with each other; a hub is a device that joins many computers in one network.

Question: [4-14] Write the causes of signal impairment and explain any one of them.

Answer: There are three causes of signal impairment: attenuation (weakening of the signal), distortion (change in the shape of the signal) and noise (unwanted signals mixing in). In attenuation, the signal becomes weak during its journey, and it is treated with an amplifier.

Question: [4-15] What is the difference between radio waves and microwaves?

Answer: Radio waves are of low frequency, can pass through obstacles and spread in all directions, and are used in AM/FM radio, TV and cordless

phones; microwaves are of higher frequency, need a line-of-sight path, and are used in mobile phones, satellites and terrestrial links.

Question: [4-16] How is the OSI model divided? Tell its layers.

Answer: The OSI model is divided into seven layers (bottom to top): Physical, Data Link, Network, Transport, Session, Presentation, Application. To remember: All People Seem To Need Data Processing.

Question: [4-17] Explain the purpose of standardisation organisations.

Answer: The purpose of standardisation organisations is to make common standards and rules so that the devices and systems of different companies can work with each other. Examples: ISO, IEEE, IETF, ITU and ANSI.

Question: [4-18] Write one advantage and one disadvantage of each topology.

Answer: Bus: cheap and easy, but if the central cable breaks, the whole network stops; ring: every computer gets the signal in turn, but if one computer fails, the whole network stops; star: easy to set up, and one computer's failure does not affect the others, but if the hub fails, the whole network stops, and more cable is needed.

Question: [4-19] Write one example each of LAN, MAN and WAN.

Answer: LAN: a school or office network; MAN: a city's cable TV network; WAN: the internet.

Question: [4-20] In which unit is the bit rate measured? Give examples.

Answer: The bit rate is measured in bits per second (bps). Examples: kilobits per second (Kbps), megabits per second (Mbps).

Question: [4-21] Which statement matches the parts of an IPv4 address?

Answer: Network and host — an IPv4 address has two parts: the network part and the host (device) part.

Question: [4-22] Which statement matches the physical address?

Answer: Data link layer — the physical (MAC) address is used at the data link layer.

Question: [4-23] Which statement matches a difference in the shape of the signal?

Answer: Distortion — a difference coming in the shape of the signal is called distortion.

Question: [4-24] Which statement matches networking that joins two networks? (As per the remaining pairing in the book's table)

Answer: WAN — a WAN joins many networks with each other over a large area.

Question: [4-25] Which statement matches the use of TCP and UDP?

Answer: Transport layer — TCP and UDP are protocols of the transport layer.

Question: [4-26] Which statement matches the OSI layer that fixes the route of the data?

Answer: The route can change — the network layer fixes the route of the data, and the route can change if needed.

Question: [4-27] Compare coaxial cable, UTP, STP and fibre-optic cable on the basis of price, data rate, installation, interference and length.

Answer: Twisted pair (UTP): cheap and easy to install, but low data rate and more interference; STP: less interference because of the shield, and faster data rate; coaxial cable: higher data rate than twisted pair and less interference, used in cable TV; fibre optic: signal by light, the highest data rate and the least interference, but expensive.

Question: [4-28] Identify the different layers of a coaxial cable.

Answer: In a coaxial cable there is a copper wire in the middle, a plastic core (insulation) around it, a mesh of metal wires (shield) outside that, and a plastic cover outermost.

Question: [4-29] What kinds of connections can be used through telephone lines?

Answer: Dial-up modem (slow connection through the telephone line), DSL modem (fast connection through the telephone line) and ISDN modem (voice, data and video through a digital network).

Question: [4-30] Make a list of the hardware needed for each topology.

Answer: Bus: central cable, terminators, NIC, connectors; ring: cable, NIC; star: hub or switch, cables, NIC. In every topology, an NIC is needed to join the computers to the network.

کمپیوٹر جو تحفظ اخلاقیات

[5-1] What is the broad term that includes different harmful software?

(a) Virus

(b) Spyware

✓ (c) Malware

(d) Adware

[5-2] Which authentication method identifies only the real user?

(a) Username password

(b) Scan code

✓ (c) Biometric

(d) PIN

[5-3] Under which protection is software mostly kept safe?

(a) Potent

✓ (b) Copyrights

(c) Trademark

(d) Logo

Question: [5-4] Professional ethics in computers are important because ...

[Note: there seems to be an error in the printed options — no option is clearly correct; the intended answer is (c), where "nahi" appears to be a misprint for "thi"]

Answer: (c) Ethics create a positive and healthy work environment — professional ethics make a work place of honesty, respect and responsibility.

[5-5] What is a free antivirus mostly like?

(a) It becomes useless after some time

✓ (b) It gives only limited service

(c) It cannot be updated

(d) It cannot be bought

[5-6] Copying some material from the internet without giving reference to its real owner is an example of what?

(a) Plagiarism

(b) Illegal use of patent

(c) Secrecy of information

✓ (d) Copyright violation

[5-7] Which malware does not copy or steal data, and so is the least harmful?

(a) Virus

✓ (b) Adware

(c) Spyware

(d) Trojan

[5-8] Which malware copies itself by itself and is not attached to any file?

(a) Virus

(b) Adware

(c) Spyware

✓ (d) Worm

Question: [5-9] What are the sources of the spread of a virus? [Note: there is an error in the printed question — all four options are real sources of virus spread; the most common source is email attachment]

Answer: All four sources are correct: a virus spreads through email attachment, internet download, flash disk/CD and CD. According to the book, the most famous source is email attachment.

[5-10] "Click the link, get 5 dollars of mobile balance" — what is this an example of?

(a) Scam

(b) Phishing

(c) Clickjacking

(d) Hacking

Question: [5-11] Why is computer security important? Write its reasons.

Answer: Computer security protects hardware, software and information from theft and damage; personal information (passwords, bank details, pictures) stays safe; the system keeps working correctly; business and government work depend on the protection of information.

Question: [5-12] Explain cyberbullying with an example.

Answer: In cyberbullying, someone is threatened, teased or defamed through a computer, mobile or the internet. Example: spreading false talk about a student on social media, or sending them insulting messages.

Question: [5-13] What is the purpose of authentication of information?

Answer: The purpose of authentication is to make sure that only the right person can reach the data or system, no one else.

Question: [5-14] What is ethical hacking?

Answer: In ethical hacking, the weaknesses of a system or network are found with permission and fixed. An ethical hacker is also called a white-hat hacker.

Question: [5-15] Your friend has become a victim of cyber harassment; what advice will you give them?

Answer: Do not answer the harasser, keep the evidence safe (screenshots), block them, tell parents or a teacher, and report to the National Response Centre for Cyber Crime (helpline 9911).

Question: [5-16] What steps should be taken to save an email account from hacking?

Answer: Keep a strong password, do not share the password with anyone, do not click unknown links or attachments, keep the antivirus updated,

and log out on a shared computer.

Question: [5-17] Why is software piracy harmful for a software developer?

Answer: Through piracy, the developer does not get the income of their hard work, the money for research on new software becomes less, and jobs are affected.

Question: [5-18] Write examples of phishing.

Answer: A false email or website asking for bank details or a password; a false message luring with a prize; a false login page that steals the password.

Question: [5-19] What is the copyright right?

Answer: Copyright is the legal right that keeps a person's own creation (book, software, writing, art) safe — another person cannot copy or use it without permission.

Question: [5-20] State the difference between virus, worm, adware and spyware according to the given criteria: level of danger, how it starts, damage to hardware/software, effect on computer speed, source of spread.

Answer: Virus: medium-high danger; starts by opening an infected file; damages files; slows the computer; spreads through infected files, email and USB. Worm: high danger; starts by itself through the network; damages the system; slows the computer a lot; spreads through the network/internet. Adware: low danger; comes with free software and advertisements; little damage, slows the system; spreads through advertisements and free software. Spyware: high danger; comes hidden in a download; steals information; slows the computer; spreads through infected websites and downloads.

Question: [5-21] Which statement matches presenting someone else's ideas as one's own?

Answer: Plagiarism — presenting someone else's writing or idea under one's own name is plagiarism.

Question: [5-22] Which statement matches the advertising software that uses pop-up windows?

Answer: Adware — adware is advertising software that shows advertisements in pop-up windows.

Question: [5-23] Which statement matches the name of a crime done through a computer system?

Answer: Cybercrime — a crime done through a computer or network is called cybercrime.

Question: [5-24] Which statement matches the secret code that identifies the user?

Answer: PIN — the PIN is that personal identification number which identifies the user.

Question: [5-25] Which statement matches the person who enters someone's computer by bypassing the password?

Answer: Cracker — a cracker is the person who enters someone's system illegally.

Question: [5-26] Which statement matches the software that saves data from damage?

Answer: Antivirus — antivirus software saves data and the system from the damage of viruses and malware.

Question: [5-27] Which points should be included in a poster on the safe use of computers and the internet?

Answer: Save your computer from viruses; avoid cyberbullying; avoid plagiarism; stay aware of cybercrime.

Question: [5-28] What should be done if someone is told of a prize on the phone and advance fee is demanded?

Answer: Do not pay the fee and end the call — demanding advance fee by the lure of a prize is fraud/scam.

Question: [5-29] What should be done if a bank account or email password is demanded in an unknown email?

Answer: Do not reply and do not tell the password — demanding personal information in an unknown email is phishing.

Question: [5-30] What should be done if Facebook or Google account details are demanded on an unknown website?

Answer: Do not write the details and close the site — demanding account details on an unknown website can lead to theft of information.

ویب ڈولپمینٹ

[6-1] Which service is responsible for public access to a website?

✓ (a) web server

(b) web hosting

(c) website

(d) web browser

[6-2] Which specific kind of website is it where users can ask questions and talk on different topics?

(a) Social site

(b) blog site

✓ (c) Forum site

(d) web browser

[6-3] A news website is specific for delivering which information?

✓ (a) News weather and current affair

(b) Promotion of business services

(c) Information of specific people

(d) Much material for video promotion

[6-4] Which code does a web browser convert, by which the document appears?

✓ (a) HTML

(b) Text

(c) doc

(d) pdf

[6-5] Which tag is used to give a new line space before or after text?

✓ (a)

(b) <hr>

(c) <p>

(d) <pre>

[6-6] Which kind of list shows the items in numbers or order?

(a) Nested list

(b) unordered list

✓ (c) Ordered list

(d) description

[6-7] Which tag is used to join (link) to another page?

✓ (a) <a> tag

(b) <u> tag

(c) <b> tag

(d) <input> tag

[6-8] Which tag is used to show the heading cell in a table?

✓ (a) <th> tag

(b) <tr> tag

(c) <td>

(d) <dt> tag

Question: [6-9] Which tag is used to show the big/main title of a web page? (Note: the <title> tag only shows the title in the browser tab; the big title on the page is shown with <h1>)

Answer: (d) <h1> — the <h1> tag shows the biggest heading on the page.

[6-10] Which attribute shows the url address of the picture in the <img> tag?

(a) target

(b) name

✓ (c) src

(d) href

Question: [6-11] State the difference between web hosting and web server.

Answer: Web hosting is the service that places a website on the internet so that all people can reach it; a web server is the computer that keeps the website's files safe and delivers them to the users.

Question: [6-12] Which steps are included in making a complete website?

Answer: Open a text editor, write the HTML code, save the file with the .html extension, and view it in the browser. For a complete site, make all the pages, join them with links, and publish them on hosting.

Question: [6-13] State the difference between <hr>, <h1>, <title> and <head>.

Answer: <hr> makes a horizontal line on the page; <h1> shows the biggest heading; <title> fixes the title of the page, which appears in the browser tab; <head> holds the extra information of the page (title, style, script) which does not appear on the page.

Question: [6-14] Identify the HTML tags used for formatting content and explain them.

Answer: <b> makes the writing bold; <i> makes the writing italic; <u> gives an underline under the writing; <pre> shows the writing as it is (preformatted); <font> fixes the colour, size and face of the font; brings the writing to the middle; <sub> puts the writing below (subscript) and <sup> above (superscript).

Question: [6-15] Tell the difference between ordered and unordered lists.

Answer: In an ordered list (<ol>), items are shown in the order of numbers or letters; in an unordered list (<ul>), items are shown with bullets, without any order.

Question: [6-16] How are pictures shown in HTML?

Answer: Pictures are shown with the <img> tag, in which the address of

the picture is written in src. Example: <img src="bridge.jpg" alt="Bridge" width="600">. <img> is an empty tag; it does not need to be closed.

Question: [6-17] How can a link go from one page to another page?

Answer: A link is made with the <a href="..."> tag. Example: Goto Google — clicking on it takes the user to the other page.

Question: [6-18] Which HTML elements are used to show data in rows and columns? Throw light on their features.

Answer: <table> makes a table; <tr> makes a row; <td> makes a data cell; <th> makes a heading cell. Features: border, width, align, cellpadding (space between the cell edge and the content), cellspacing (space between cells); cell features: align, valign, colspan, rowspan.

Question: [6-19] Is it possible for the content of one page to appear together with another page? How?

Answer: Yes, it is possible with frames. <frameset> divides the page into rows/columns, and in each part another page is shown with <frame src="...">.

Question: [6-20] Write the names of famous text editors and web designing tools that help in making a website.

Answer: Notepad, Microsoft FrontPage, Adobe Dreamweaver, WordPress, Wix, Microsoft Visual Studio, Adobe XD.

Question: [6-21] Which statement matches e-commerce?

Answer: Business website — in e-commerce, business is done through the internet.

Question: [6-22] Which statement matches the <head> tag?

Answer: Containing the extra information and styling features of the web page — <head> holds things like title, style and script.

Question: [6-23] Which statement matches the <ul>, <ol> and <dl>

tags?

Answer: Arranging information in the form of lists — with these tags, information is shown in the form of lists.

Question: [6-24] Which statement matches the <tables> tag?

Answer: Arranging information in rows and columns — with a table, information is shown in rows and columns.

Question: [6-25] Which statement matches the <img> tag?

Answer: Adding pictures and infographics on the web page — with the <img> tag, pictures are placed on the page.

Question: [6-26] Which statement matches a hyperlink?

Answer: Adding navigation support in web pages — with a hyperlink, the user can go from one page to another page.

Question: [6-27] Make the home.html page: file name home.html, page title "unit 6 - activity home", heading "input and out devices", input devices, output devices, and "input device" link to input.html, "output device" link to output.html.

Answer: <html><head><title>unit 6 - activity home</title></head><body><h1>input and out devices</h1> Input Devices Output Devices input device
output device</body></html>

Question: [6-28] Make the inputs.html page: page title "unit 6 - activity - input device", a two-column page; in the left column the names of different input devices, in the right column the pictures of those devices (200 pixels).

Answer: <html><head><title>unit 6 - activity - input device</title></head><body><table><tr><td>Keyboard
Mouse
Scanner</td><td><img src="keyboard.jpg" width="200"><img

src="mouse.jpg" width="200"><img src="scanner.jpg" width="200"></td></tr></table></body></html>

Question: [6-29] Make the outputs.html page: a two-column page; in the left column the names of different output devices, in the right column the pictures of those devices (200 pixels).

Answer: <html><head><title>unit 6 - activity - output device</title></head><body><table><tr><td>Monitor
Printer
Speaker</td><td><img src="monitor.jpg" width="200"><img src="printer.jpg" width="200"><img src="speaker.jpg" width="200"></td></tr></table></body></html>

ڈیٹا بیس سسٹم جو تعارف

[7-1] Which of the following Microsoft programs is a DBMS?

(a) MS Word

(b) MS Excel

(c) Oracle

✓ (d) MS Access

[7-2] What is the basic limit of a flat file system?

(a) It is complex

✓ (b) It keeps data in a single file

(c) It is very pro

(d) It does not support MS Access

[7-3] What is a whole row of one student in the students' table called? (The printed question is unclear; it is read in the sense of a row)

✓ (a) Record

(b) Field

(c) MS data type

(d) Entity

[7-4] What is one piece of information (such as the name) of one student in the students' table called? (The printed question is unclear; it is read in the sense of one piece of information)

(a) Record

✓ (b) Field

(c) MS data type

(d) Entity

[7-5] In what form is data kept in a relational database? (The printed question is unclear)

(a) Table

(b) Field

(c) Record

(d) MS relation

[7-6] In which data type do the students' names come in a field?

(a) Whole number

(b) Float

✓ (c) String

(d) MS Boolean

[7-7] What is a field that uniquely identifies each record called?

✓ (a) Primary key

(b) Secondary key

(c) Foreign key

(d) Super key

[7-8] Data redundancy means ... of data.

✓ (a) Duplication

(b) Quality of data

(c) Size of data

(d) Type of data

[7-9] Which relation joins one record of one entity with one record of another entity?

✓ (a) One-to-one relation

(b) One-to-many relation

(c) Many-to-one relation

(d) Many-to-many relation

[7-10] Which shape is used to show attributes in an ERD?

(a) Diamond

(b) Octagon

✓

(c) Rectangle

(d) Oval

Question: [7-11] Why is a database management system preferred over a flat file system?

Answer: In a DBMS, many users can use the data at the same time, data stays safe, there is no duplication, online access is possible, and data is easily updated. In a flat file system, there is only one user, no security, and the problem of duplication.

Question: [7-12] Discuss the use of databases in business with examples.

Answer: In business, databases keep customer records, stock, sales, employees' salaries and accounts. Examples: a shop keeps records of customers and stock; a bank keeps records of accounts.

Question: [7-13] What is the difference between Datasheet View and Design View?

Answer: In Datasheet View, data appears in rows and columns, and data can be entered or changed in it; in Design View, the structure of the table (field names, data types, properties) appears, and the table is designed/changed there.

Question: [7-14] Why is the data type fixed in each field?

Answer: So that only the correct kind of data is entered in the field, there are no mistakes, space is saved, and searching/sorting is correct. Example: in a Number field, only numbers can be entered.

Question: [7-15] What is the difference between a table and a database?

Answer: A table is one set of arranged rows and columns about one topic; a database is a collection of many related tables (and queries, forms, reports).

Question: [7-16] Explain entity, attributes and relation with one example each.

Answer: Entity: an independent thing, such as a student; attribute: a quality of the entity, such as the student's name, class; relation: the link between entities, such as a student belongs to a class.

Question: [7-17] What is the difference between primary and foreign keys?

Answer: A primary key uniquely identifies each record in its own table (it is never repeated, never empty); a foreign key is the primary key of another table, used in this table so that the two tables can be joined.

Question: [7-18] Write the advantages of using relations in a database.

Answer: Duplication of data ends; data stays consistent and correct; less space is used; combined information is easily obtained from many tables.

Question: [7-19] Discuss the importance of ERD (in business).

Answer: In an ERD, entities, attributes and relations are shown in the form of a diagram; with it, the database is correctly designed before making it, mistakes are avoided, time is saved, and the ERD works as documentation/reference.

Question: [7-20] Describe the given ER diagram: entities — Company, Advertisement, Candidate, Interview; relations — issued, applied for, taken, faced.

Answer: The company issues the advertisement; the candidate applies for the advertisement; the company takes the interview; the candidate faces the interview. One company can issue many advertisements (one-to-many), and one candidate can face many interviews.

Question: [7-21] Which statement matches the primary key?

Answer: Always a unique value in the field — the value of a primary key is always unique.

Question: [7-22] Which statement matches whole-number data?

Answer: Numbers without a decimal point — whole numbers are numbers without a decimal.

Question: [7-23] Which statement matches a relation?

Answer: A link between two tables — a relation joins two tables with each other.

Question: [7-24] Which statement matches a field?

Answer: Attribute — a field is also called an attribute.

Question: [7-25] Which statement matches data?

Answer: Data in the form of simple facts — data are simple facts that have not been arranged.

Question: [7-26] Which statement matches the flat file system? (As per the remaining pairing in the book's table)

Answer: A system without a data dictionary — there is no data dictionary in a flat file system.

Question: [7-27] Which statement matches an entity?

Answer: A table with its attributes — an entity is in the form of a table with its attributes.